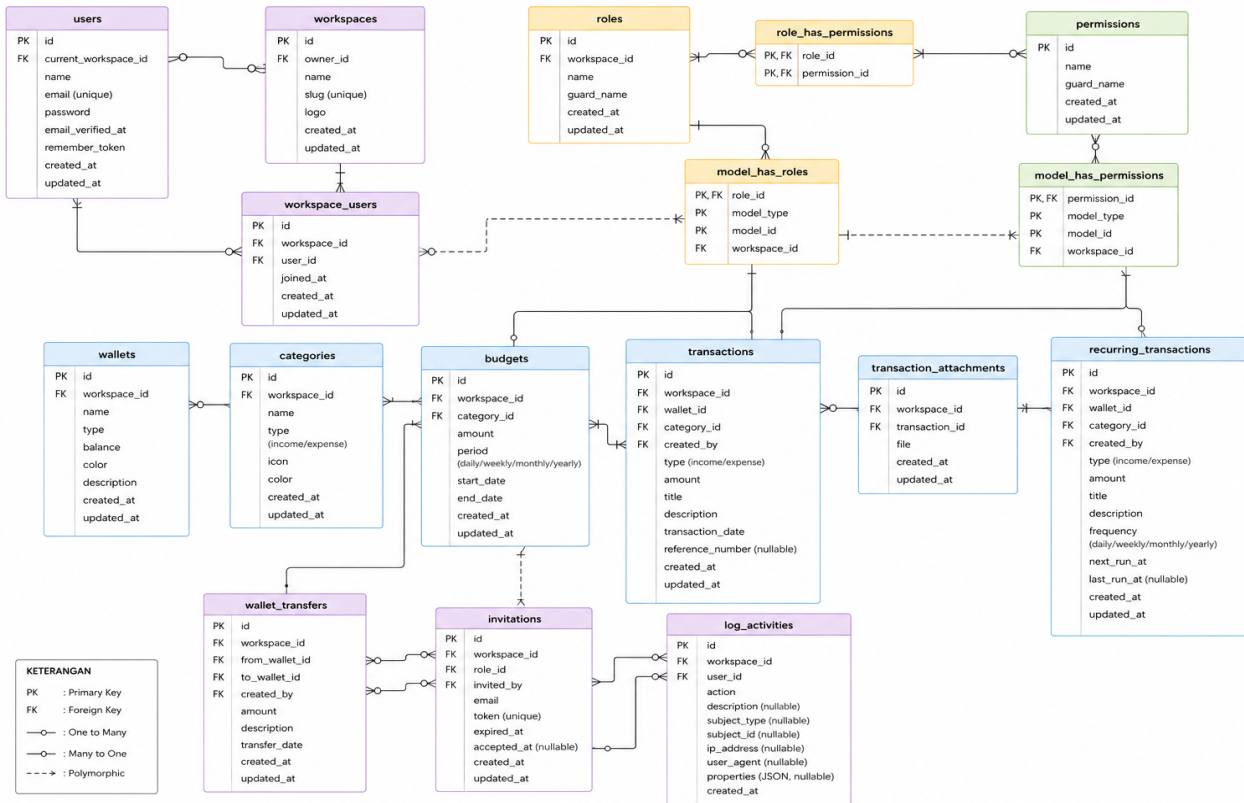


Cashflow Tracker SaaS Multi-Tenant RBAC Architecture

Database Architecture, System Flow, and Table Explanation

This document explains the architecture design for a SaaS-based Cashflow Tracker application using Multi-Tenant architecture and RBAC (Role-Based Access Control).

ERD Diagram



System Flow

1. User Registration

- User registers to the application.
- System automatically creates a workspace.
- User becomes the owner of the workspace.

2. Workspace Management

- Owner can invite team members.
- Members can join multiple workspaces.
- Each workspace has isolated data.

3. RBAC Authorization

- Permissions are global and defined by the system.
- Roles are created per workspace.
- Owners can create custom roles and assign permissions.

4. Transaction Management

- Users can create income and expense transactions.
- Transactions belong to wallets and categories.

- All business data is scoped using workspace_id.

Core Architecture

Multi Tenant

All business data belongs to a workspace. Every business table contains workspace_id to isolate tenant data.

RBAC

Role-Based Access Control allows flexible permission management per workspace.

Scalable SaaS Design

The architecture supports:

- Multi company support
- Invite members
- Team collaboration
- Subscription plans
- Audit logs
- Future enterprise scaling

Database Table Explanation

Table	Description
users	Stores login account data.
workspaces	Main tenant/company/workspace entity.
workspace_users	Relation between users and workspaces.
roles	Workspace-scoped RBAC roles.
permissions	Global system permissions.
wallets	Stores wallets/accounts/cash balances.
categories	Transaction categories.
budgets	Budget limit per category.
transactions	Income and expense records.
transaction_attachments	Transaction file attachments.
recurring_transactions	Automatic recurring transactions.
wallet_transfers	Transfer between wallets.
invitations	Workspace invitation system.
log_activities	User activity and audit logs.

This architecture is production-ready for building a scalable SaaS Cashflow Tracker application using Laravel, Vue.js, PostgreSQL, and Spatie Permission with Multi-Tenant RBAC.